

Justin Q. Li

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Education

University of North Carolina at Chapel Hill – Chapel Hill, NC

May 2026

National University of Singapore – Singapore, Singapore

May 2025

Bachelor of Computer Science BS

Coursework: Data Structures and Algorithms, Systems Fundamentals, Modern Web Programming, Cryptography, Machine Learning, Artificial Intelligence, Discrete Structures, Compression and Streaming

Honors and Awards: Covenant Scholar (Full Tuition), Questbridge Scholar, Susan Wallace Scholar, Gilman International Scholar

Skills and Technical Tools

Languages: Python, Java, React.js, Vue.js, Node.js, Ruby on Rails, Swift, MySQL, PostgreSQL, Javascript, C++, C#, C, R

Technologies: Pytorch, Tensorflow, Pandas, Angular, Bootstrap, Spring Boot, Terraform, Microsoft Azure, AWS, Flask, JavaFX, RESTful APIs

Proficiencies: Software development, Mobile development, Cybersecurity, Machine learning, Data Science, UI/UX design, DevOps

Experience

Software Engineer (Compute Automation) | Wells Fargo – Charlotte, NC

Jun 2025 – Aug 2025

- Spearheaded a 20-member cross-functional initiative to enhance system delivery velocity, yielding a **99.6%** success rate across CTO.
- Engineered automated CI/CD (Continuous Integration/Continuous Deployment) pipelines using GitHub Actions and Ansible Playbooks to streamline deployments and enforce compliance, reducing manual errors by **40%**.
- Built a Python-based patch identification service using Tachyon LLM to cluster seed URLs and guide product scraping, accelerating patch discovery and resolution.

Software Engineer (Full-stack Web) | Ideas United – Atlanta, GA

Jun 2024 – Aug 2024

- Developed **10+** client projects using Vue.js, Nuxt.js, Ruby on Rails, and PostgreSQL, resulting in **50+** partner UGC submissions and **4000+** monthly internal user site visits.
- Employed DigitalOcean cloud infrastructure, real-time communication using AnyCable-GO, Gemset for environment management, Maildump for email testing, serializers for efficient data handling, and GitLab for version control and CI/CD.

iOS Developer | Porch – Chapel Hill, NC

Aug 2023 – May 2026

- Engineered **15+** functional iOS applications using Swift, Xcode, Git, Figma, Twilio, Splunk, and Firebase while implementing MVVM architecture, integrating APIs, and following Agile methodologies.
- Developed an iOS app for message blasting, volunteer organization, and data collection, streamlining operations for PORCH communities, contributing to PORCH's efforts in generating **\$6.5** million in hunger relief.

Web Developer | Board of Regents of the University System of Georgia – Atlanta, GA

May 2021 – Jul 2021

- Developed and maintained University System of Georgia domains, implementing image compression, browser caching, and code minification performance optimization strategies, achieving a **25%** reduction in load times.

Founder | SoleLiKicks – Ozark, AL

Jan 2018 – Jan 2024

- Founded SoleLiKicks, a **five-figure** retail arbitrage service with a concentration in footwear.

Projects

GreenerModel — Energy-Efficient LLM Routing System | Python, PyTorch, FastAPI, DistilBERT, LightGBM, Redis, Prometheus

- Built an intelligent routing system dynamically selecting the smallest sufficient language model per request, reducing energy consumption by **60%** and compute costs while maintaining **95%+** answer quality through confidence-based escalation.

Arbor — Product Deployment Automation Framework | Ansible, Python, Spring Boot, Helix, Splunk, ServiceNow, BigPanda, Apache Kafka

- Automated middleware delivery across teams using Helix Orchestration and Dockerized packages with edge-case detection.
- Developed a Java API to parse deployment errors using regex and publish to Kafka topics linked to corrective Ansible playbooks, reducing mean time-to-resolution by **40%**.

MacroInsightML — Machine Learning Analysis of Environmental & Clinical Risk Factors | Python, Pandas, Scikit-Learn, Anthropic API

- Built a data pipeline to model national economic performance across **1,900+** country-year records using regression and classification.
- Implemented linear, ridge, logistic, and SGD-based models with full preprocessing, evaluation, and LLM-assisted result interpretation, achieving up to $R^2 = 0.47$ and **85%+** classification accuracy.

Codec Benchmarking Platform — 3D Compression Evaluation Engine | Draco, MPEG G-PCC, MPEG V-PCC, PCL

- Built an automated benchmarking system to evaluate Draco, MPEG G-PCC, MPEG V-PCC, and PCL across **1,211** real-world 3D point cloud models, measuring compression ratio, encoding/decoding latency, and PSNR-based reconstruction fidelity.

ByteBuddy — NC Triangle Calorie Tracker | Swift, SwiftUI, SwiftSoup, WebKit, AlamoFire, Firebase

- Spearheaded the development of a nutrition assistant mobile app, implementing features such as meal logging, personalized nutrition reports, and real time nutrition information for **1200+** items across **3** campuses.